

Part2

```
use("shopDB")
```

```
db.products.insertMany([
```

```
  { _id: 1, name: "Laptop", price: 999.99, category: "Electronics", tags: ["portable",  
"wireless"], stock: 15 },
```

```
  { _id: 2, name: "Smart Watch", price: 199.99, category: "Electronics", tags: ["wearable",  
"wireless"], stock: 30 },
```

```
  { _id: 3, name: "Desk Chair", price: 149.99, category: "Furniture", tags: ["ergonomic",  
"office"], stock: 8 },
```

```
  { _id: 4, name: "Coffee Maker", price: 49.99, category: "Appliances", tags: ["kitchen",  
"morning"], stock: 25 },
```

```
  { _id: 5, name: "Headphones", price: 79.99, category: "Electronics", tags: ["audio",  
"wireless"], stock: 50 }  
])
```

```
db.orders.insertMany([
```

```
  { _id: 1, customer: "Alice", date: new Date("2024-01-15"), items: [{ productId: 1, qty: 1  
}, { productId: 5, qty: 2 }], total: 1159.97, status: "delivered" },
```

```
  { _id: 2, customer: "Bob", date: new Date("2024-02-03"), items: [{ productId: 2, qty: 1 }],  
total: 199.99, status: "shipped" },
```

```
  { _id: 3, customer: "Alice", date: new Date("2024-02-20"), items: [{ productId: 3, qty: 1  
}, { productId: 4, qty: 2 }], total: 249.97, status: "pending" },
```

```
  { _id: 4, customer: "Carol", date: new Date("2024-03-10"), items: [{ productId: 5, qty: 3  
}], total: 239.97, status: "pending" }  
])
```

```
Atlas atlas-10lj3g-shard-0 [primary] test> use("shopDB")  
switched to db shopDB  
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.insertMany([  
|   { _id: 1, name: "Laptop", price: 999.99, category: "Electronics", tags: ["portable", "wireless"], stock: 15 },  
|   { _id: 2, name: "Smart Watch", price: 199.99, category: "Electronics", tags: ["wearable", "wireless"], stock: 30 },  
|   { _id: 3, name: "Desk Chair", price: 149.99, category: "Furniture", tags: ["ergonomic", "office"], stock: 8 },  
|   { _id: 4, name: "Coffee Maker", price: 49.99, category: "Appliances", tags: ["kitchen", "morning"], stock: 25 },  
|   { _id: 5, name: "Headphones", price: 79.99, category: "Electronics", tags: ["audio", "wireless"], stock: 50 }  
| ])  
{  
  acknowledged: true,  
  insertedIds: { '0': 1, '1': 2, '2': 3, '3': 4, '4': 5 }  
}
```

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.insertMany([  
|   { _id: 1, customer: "Alice", date: new Date("2024-01-15"), items: [{ productId: 1, qty: 1 }, { productId: 5, qty: 2 }], total: 11  
59.97, status: "delivered" },  
|   { _id: 2, customer: "Bob", date: new Date("2024-02-03"), items: [{ productId: 2, qty: 1 }], total: 199.99, status: "shipped" },  
|   { _id: 3, customer: "Alice", date: new Date("2024-02-20"), items: [{ productId: 3, qty: 1 }, { productId: 4, qty: 2 }], total: 24  
9.97, status: "pending" },  
|   { _id: 4, customer: "Carol", date: new Date("2024-03-10"), items: [{ productId: 5, qty: 3 }], total: 239.97, status: "pending" }  
| ])  
{ acknowledged: true, insertedIds: { '0': 1, '1': 2, '2': 3, '3': 4 } }  
Atlas atlas-10lj3g-shard-0 [primary] shopDB>
```

Part 3 — CRUD

3.1 Find

1) db.products.find()

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.find()
[
  {
    _id: 1,
    name: 'Laptop',
    price: 999.99,
    category: 'Electronics',
    tags: [ 'portable', 'wireless' ],
    stock: 15
  },
  {
    _id: 2,
    name: 'Smart Watch',
    price: 199.99,
    category: 'Electronics',
    tags: [ 'wearable', 'wireless' ],
    stock: 30
  },
  {
    _id: 3,
    name: 'Desk Chair',
    price: 149.99,
    category: 'Furniture',
    tags: [ 'ergonomic', 'office' ],
    stock: 8
  },
  {
    _id: 4,
    name: 'Coffee Maker',
    price: 49.99,
    category: 'Appliances',
    tags: [ 'kitchen', 'morning' ],
    stock: 25
  },
]
```

2) db.products.find({ category: "Electronics" }, { name: 1, price: 1, _id: 0 })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.find({ category: "Electronics" }, { name: 1, price: 1, _id: 0 })
[
  { name: 'Laptop', price: 999.99 },
  { name: 'Smart Watch', price: 199.99 },
  { name: 'Headphones', price: 79.99 }
]
```

3) db.products.find({ price: { \$gt: 50 } }).sort({ price: -1 })

```
[
  {
    _id: 1,
    name: 'Laptop',
    price: 999.99,
    category: 'Electronics',
    tags: [ 'portable', 'wireless' ],
    stock: 15
  },
  {
    _id: 2,
    name: 'Smart Watch',
    price: 199.99,
    category: 'Electronics',
    tags: [ 'wearable', 'wireless' ],
    stock: 30
  },
  {
    _id: 3,
    name: 'Desk Chair',
    price: 149.99,
    category: 'Furniture',
    tags: [ 'ergonomic', 'office' ],
    stock: 8
  },
  {
    _id: 5,
    name: 'Headphones',
    price: 79.99,
    category: 'Electronics',
    tags: [ 'audio', 'wireless' ],
    stock: 50
  }
]
```

3.2 Insert

`db.products.insertOne({ _id: 6, name: "Bluetooth Speaker", price: 49.99, category: "Electronics", tags: ["speaker", "audio"], stock: 60 })`

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.insertOne({ _id: 6, name: "Bluetooth Speaker", price: 49.99, category: "Electronics", tags: ["speaker", "audio"], stock: 60 })
{ acknowledged: true, insertedId: 6 }
```

3.3 Update

1) `db.products.updateMany({ category: "Electronics" }, { $mul: { price: 1.10 } })`

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.updateMany({ category: "Electronics" }, { $mul: { price: 1.10 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
```

2) `db.products.updateOne({ name: "Smart Watch" }, { $push: { tags: "best-seller" } })`

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.updateOne({ name: "Smart Watch" }, { $push: { tags: "best-seller" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

3.4 – Delete

1) db.products.deleteOne({ _id: 6 })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.deleteOne({ _id: 6 })
{ acknowledged: true, deletedCount: 1 }
```

2) db.products.insertOne({ _id: 7, name: "Old Cable", price: 4.99, category: "Accessories", tags: [], stock: 3 })

db.products.deleteMany({ stock: { \$lt: 10 } })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.insertOne({ _id: 7, name: "Old Cable", price: 4.99, category: "Accessories",
tags: [], stock: 3 })
{ acknowledged: true, insertedId: 7 }
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.deleteMany({ stock: { $lt: 10 } })
{ acknowledged: true, deletedCount: 2 }
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

Part 4: Querying with Operators

1. db.products.find({ price: { \$gte: 20, \$lte: 100 } })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.find({ price: { $gte: 20, $lte: 100 } })
[
  {
    _id: 4,
    name: 'Coffee Maker',
    price: 49.99,
    category: 'Appliances',
    tags: [ 'kitchen', 'morning' ],
    stock: 25
  },
  {
    _id: 5,
    name: 'Headphones',
    price: 87.989,
    category: 'Electronics',
    tags: [ 'audio', 'wireless' ],
    stock: 50
  }
]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

2. db.orders.find({ status: { \$in: ["pending", "shipped"] } })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.find({ status: { $in: ["pending", "shipped"] } })
[
  {
    _id: 2,
    customer: 'Bob',
    date: ISODate('2024-02-03T00:00:00.000Z'),
    items: [ { productId: 2, qty: 1 } ],
    total: 199.99,
    status: 'shipped'
  },
  {
    _id: 3,
    customer: 'Alice',
    date: ISODate('2024-02-20T00:00:00.000Z'),
    items: [ { productId: 3, qty: 1 }, { productId: 4, qty: 2 } ],
    total: 249.97,
    status: 'pending'
  },
  {
    _id: 4,
    customer: 'Carol',
    date: ISODate('2024-03-10T00:00:00.000Z'),
    items: [ { productId: 5, qty: 3 } ],
    total: 239.97,
    status: 'pending'
  }
]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

3. db.products.find({ tags: "wireless" })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.products.find({ tags: "wireless" })
[
  {
    _id: 1,
    name: 'Laptop',
    price: 1099.989,
    category: 'Electronics',
    tags: [ 'portable', 'wireless' ],
    stock: 15
  },
  {
    _id: 2,
    name: 'Smart Watch',
    price: 219.98900000000003,
    category: 'Electronics',
    tags: [ 'wearable', 'wireless', 'best-seller' ],
    stock: 30
  },
  {
    _id: 5,
    name: 'Headphones',
    price: 87.989,
    category: 'Electronics',
    tags: [ 'audio', 'wireless' ],
    stock: 50
  }
]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

4. db.orders.find({ customer: { \$ne: "Alice" } })

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.find({ customer: { $ne: "Alice" } })
[
  {
    _id: 2,
    customer: 'Bob',
    date: ISODate('2024-02-03T00:00:00.000Z'),
    items: [ { productId: 2, qty: 1 } ],
    total: 199.99,
    status: 'shipped'
  },
  {
    _id: 4,
    customer: 'Carol',
    date: ISODate('2024-03-10T00:00:00.000Z'),
    items: [ { productId: 5, qty: 3 } ],
    total: 239.97,
    status: 'pending'
  }
]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

Part 5: Aggregation Pipeline

1. db.orders.aggregate([{ \$group: { _id: null, totalRevenue: { \$sum: "\$total" } } }])

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.aggregate([ { $group: { _id: null, totalRevenue: { $sum: "$total" } } } ])
[ { _id: null, totalRevenue: 1849.9 } ]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

2. db.orders.aggregate([{ \$group: { _id: "\$customer", avgOrderValue: { \$avg: "\$total" } } }])

```
Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.aggregate([ { $group: { _id: "$customer", avgOrderValue: { $avg: "$total" } } } ])
[
  { _id: 'Carol', avgOrderValue: 239.97 },
  { _id: 'Alice', avgOrderValue: 704.97 },
  { _id: 'Bob', avgOrderValue: 199.99 }
]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |
```

3.

```

db.orders.aggregate([
  { $unwind: "$items" },
  { $group: { _id: "$items.productId", totalQty: { $sum: "$items.qty" } } },
  { $lookup: { from: "products", localField: "_id", foreignField: "_id", as: "productInfo" } },
],
  { $unwind: "$productInfo" },
  { $project: { productName: "$productInfo.name", totalQty: 1 } },
  { $sort: { totalQty: -1 } }
])

```

```

Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.aggregate([
|   { $unwind: "$items" },
|   { $group: { _id: "$items.productId", totalQty: { $sum: "$items.qty" } } },
|   { $lookup: { from: "products", localField: "_id", foreignField: "_id", as: "productInfo" } },
|   { $unwind: "$productInfo" },
|   { $project: { productName: "$productInfo.name", totalQty: 1 } },
|   { $sort: { totalQty: -1 } }
| ])
|
| [
|   { _id: 5, totalQty: 5, productName: 'Headphones' },
|   { _id: 4, totalQty: 2, productName: 'Coffee Maker' },
|   { _id: 2, totalQty: 1, productName: 'Smart Watch' },
|   { _id: 1, totalQty: 1, productName: 'Laptop' }
| ]
Atlas atlas-10lj3g-shard-0 [primary] shopDB> |

```

4.

```

db.orders.aggregate([
  { $group: { _id: { year: { $year: "$date" }, month: { $month: "$date" } }, orderCount: { $sum: 1 } } },
  { $sort: { "_id.year": 1, "_id.month": 1 } }
])

```

```

Atlas atlas-10lj3g-shard-0 [primary] shopDB> db.orders.aggregate([
|   { $group: { _id: { year: { $year: "$date" }, month: { $month: "$date" } },
|   orderCount: { $sum: 1 } } },
|   { $sort: { "_id.year": 1, "_id.month": 1 } }
| ])
|
| [
|   { _id: { year: 2024, month: 1 }, orderCount: 1 },
|   { _id: { year: 2024, month: 2 }, orderCount: 2 },
|   { _id: { year: 2024, month: 3 }, orderCount: 1 }
| ]
Atlas atlas-10lj3g-shard-0 [primary] shopDB>

```

Part 6: Schema Design – Embedding vs. Referencing

Scenario A

It's better to use Referencing

Comments need to be read not only along with the post but also analyzed separately. Therefore, it's more convenient to store them in a separate collection and link them via postId.

Scenario B

It's better to use Referencing

A product can have hundreds of reviews, which would make the document too large. Also, users search for their reviews, so it's better to store them separately and link them via productId.

Part 7: Indexing Concept

1. Instead of searching through all the documents in the orders collection, MongoDB immediately finds the required records by the customer field.

2.

Category

Often used for filtering , speeds up category searches

Name

Used to search for a specific product, speeds up searches by name